Participant Information Leaflet

Evaluating Plausibility in VR Beach Volleyball with varying levels of Game and Crowd Population

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Invitation

Hi! If you are reading this document, it means you have been invited to participate in a research study inspecting how the population of and around a beach volleyball environment in Virtual Reality (=VR) affects immersion. First of all thank you for taking the time and interest to read this!

Before you commit to participating, there are a few questions you might have about the study and your role in it, which this document aims to answer to the best of my (the researcher's) ability.

What is the Purpose of the Study?

In an effort to make virtual environments more believable and immersive for users, there is a lot of research needed to establish good practises regarding what parameters can affect users' experiences both positively or negatively.

Considering it is by now an established market for entertainment as well as for professional use, I believe it to be useful to determine how, in a VR Sports application, specifically Beach Volleyball, the 'naturality' of teammates and opponents (how different does it feel to play with/against a CPU vs another Human player), as well as the realism of crowds on the sidelines at various stages, affects a player's perception of the game.

This study is conducted towards my master's thesis to complete my M.Sc. Computer Science degree specialised on Augmented and Virtual Reality technologies.

Why have I been invited to take part?

You are being invited because you have expressed interest in participating! Thank you for that!

Your participation is conditional to your fulfilling the selection criteria, which are that you are a legal adult in Ireland (aged 18 or older), and able to provide informed consent without the requirement of external approval. If you do not meet these criteria you may not participate in the study.

Do I have to take part?

Nope! Your participation is entirely up to whether you want to do it or not, including after you signed the consent form. If you decide it's not for you, that's absolutely fine :)

What will happen if I participate?

a) Appointment setting
This document should have been sent alongside a few proposed date-time slots. Once we have agreed an appointment for the experiment, we will meet at a pre-agreed place (most likely the SCSS Computer Labs at 11 Leinster Street South, Dublin) to access the staging area for the trials.

* Considering you will be playing a game requiring physical activity (VR Beach Volleyball), I recommend attending in clothes you are comfortable moving and sweating in. There may not be any changing rooms, though you will always be able to get changed in the restrooms. Arrangements will be made to have water and a clean towel available on-site as well, though the latter would also be best brought by yourself.

** Due to the nature of the trials, they may be organised to host up to two participants at the same time, including yourself. You will always be informed if that is the case, and offered to reschedule if you are not comfortable with such arrangements.

In total, please expect a time frame of 60-90 minutes to take part in this experiment.

b) Information and Consent Form (~5 min)
There, you will be asked to sign a consent form (which should also have been sent with this leaflet, please reach out to me if that is not the case), with the possibility to ask further questions to the researcher directly prior to signature. This consent is retractable at any time after your participation, should you request its invalidation in writing (e.g. via email) to the researcher.

c) Setup and Tutorial (~15 min)
Once you are ready to take part, the first step of the actual trials will be getting familiar with the technology and software. There will be a short tutorial to get acquainted with the way the game works, with the researcher at hand to help set up the system properly and navigate to the application. The tutorial phase will be set up to take approximately 15 minutes.

d) Gameplay (30-45 min)
Once the tutorial is completed, you will participate in a few short matches with various

configurations. For the first two matches, you will always play one game involving only computer-controlled players (CPU), and one with the other participant or the researcher in their absence. Between points in the matches, you will be encouraged to tweak the crowd population to your liking, but should you feel comfortable with a certain preset, you will be able to remain in this configuration. The gameplay phase is expected to take 30-45 minutes.

e) Questionnaire (10-15 min)
After you have played a certain amount, or decided to end the experiment early, you will be directed to complete a questionnaire relating to your experience, which should take an estimated 10-15 minutes to answer.

What are the benefits of participating?

There is no direct benefit from participation, beyond the opportunity to play a game in VR, and the pride of participating in a research project :)

What risks to me are involved in participation?

There is a low risk of physical discomfort when using VR hardware, which may include eye strain, dizziness, or motion sickness. Should you experience any such discomfort, the session will be immediately interrupted, and you may remove the headset. There will be **no consequences** should you choose not to continue with the trials after a such pause.

Another risk associated with VR sports is the mismatch between the digital and physical space, which could manifest in accidental contact with the physical environment. As such you will be encouraged to *minimise your movements*, although the environment will have been curated to make the likelihood of such accidents as low as possible, on top of other measures built into the hardware to avoid such scenarios.
However, you will be *discouraged* from performing moves that involve either upsetting your balance (e.g. jumping, diving, or leaning too far in one direction), or happen too suddenly and may cause harm on contact (e.g. spiking the ball strongly).

One last risk to your privacy to consider is the identifiability of your movements, which has been proven able to lead to a so-called "motion signature" widely recognisable from surprisingly little motion data, as well as the potential for inference of personal data such as height from factors needed for application logic (headset and controller position from the ground/each other for example).
- The application you will be playing is made by the researcher, and **commits to not processing** such data beyond direct application logic in any way.
- The only risks left to such data's availability is safety in transit for the networked part of the application, which is *secured by an industry-grade game development framework*, and

- collection mechanisms embedded into the hardware, which is provided by **Meta Technologies** (formerly Facebook). The researcher cannot guarantee such data is not sent to the hardware provider; the best mitigation I can offer in that regard is that you will not be asked to log into anything on the VR device, meaning that your motion data will confuse the researcher's profile rather than lead to your direct identification.

Will my participation in this study be kept confidential?

Yes absolutely. There is very little data that is collected that can lead to your participation being conclusively proven, and that is kept entirely separate from anything that will be shown in the paper produced as a result of this study (which will probably not be published either).

In terms of personally identifiable data, your name and email address will be collected to enable contact channels to set up the date and time for the experiment, as well as exchange down the line should any problems arise (e.g. to identify what data to delete should you request its erasure). This information will be kept entirely confidential throughout the length of the study (until including August 26th, 2026), and kept separate from any other data in the study, linking the two through an abstract ID.

Towards the study, your approximate age (in groupings of 18-24, 25-40 etc.) as well as your gender will be requested in the questionnaire to illustrate demographic data, e.g. to outline potential biases or explain interesting differences. You will be asked not to log into the survey's service (Google or Microsoft) for best anonymity, as it will not be made a requirement for privacy reasons.

The questionnaire also includes open text fields, which you will be asked to fill as anonymously as possible. All answers will be post-processed for full anonymisation either way, in order to be able to quote from these answers to illustrate general impressions of the application.

On the application layer, the only data extracted for research will be your choice of adjustment for the crowd population (including a history of states throughout the matches), as well as your choice of playing with or without Human involvement after the 2nd match.
In enabling the application to work as intended, the position and orientation of the VR headset and the two controllers will be used to map your interactions with the ball, as well as the buttons pressed on the controller to estimate hand gestures - both for ball interactions and for avatar realism for other Humans. As mentioned previously in the risks section, such data will not be stored in any way to avoid enabling a motion profile.

How is my data stored and protected?

The list of names and email addresses will be stored in a password-protected file, most likely in .csv format.

Questionnaire responses will be intermediately stored on the website it is hosted on (again, most likely Microsoft or Google Forms) until it is post-processed for pseudonymisation to satisfaction, after which it will be stored in a separate .csv file, with a unique identifier serving as a link between the two files.

Data collected from the application (as described above) will be written to a file on the headset during runtime, which will then be extracted to my laptop for processing and storage after the session has ended. The files on headset will be deleted after the move, and the resulting files will be added to a password-protected database, which will be used for the study as well.

The files above will be stored on my private NextCloud storage, which puts in place many security measures to safeguard files both in storage (through file encryption in AES-256 format) and in transit (using connections secured through TLS/SSL using reputable handshake, encryption and signature algorithms).

How long will my data be kept?

All personally identifiable data (names, email addresses) will be deleted to the best of the researcher's ability as soon as the study ends on August 26th, 2026, and no later than a month after this date. All data in general will be fully anonymised as soon as it is practically feasable without impacting the study's progress.

What will happen to the results of the study?

The information from this study will feature in a master's dissertation, with no current plans of publication. This information may be used to motivate further research projects to build upon the findings.

In any case, your participation will be "hidden" behind aggregations of quantitative data, or quotes from your open text responses, editorialised for pseudonymisation, or even anonymisation if possible should they prove to be helpful in illustrating impressions of the system.

Who is organising and funding the study?

The study is conducted by Joshua O'Donnell towards a Master's Dissertation at Trinity College Dublin. The project is not externally funded.

Who do I contact for further information?

For any further questions about the study specifically, please contact

Joshua O'Donnell MSc Computer Science - AVR under the email address jodonne5@tcd.ie

For any higher concerns you wish to bring independently of the researcher, please contact the relevant authority at Trinity College Dublin through the School of Computer Science and Statistics.

Thanks again for your interest, and I hope this huge wall of text was helpful in cementing your intention to proceed with the study!

Data Protection Information

Data Controller: Trinity College Dublin
Data Protection Officer: Data Protection Officer, Secretary's Office, Trinity College Dublin, Dublin 2, Ireland; email: dataprotection@tcd.ie; website www.tcd.ie/privacy

What is the legal basis to use my personal data?

Personally Identifiable information will only be used for communication if pressing matters (e.g. data leaks, correspondence w.r.t. legal rights and obligations) occur. The legal basis for processing your personal data is Article 6(1)(e) of the EU General Data Protection Regulation (GDPR).

What are my rights to my data?

Disclaimer: I am not a lawyer; The following section relates to my understanding of the legal guidelines this study is subject to. For any legally binding explanations or operation please consult the cited legislation or contact a legal professional.

This research is conducted in Ireland, a member state of the European Union, meaning all personal data collected towards this study is subject to the EU's GDPR and Irish Data Protection Legislations. For information purposes, among others, you have the right to:

- be informed of what data is collected, how it is collected, and the purpose it is collected towards,
- access a copy of your personal data and information on how it is protected,
- rectify inaccurate data (e.g. if your name is stored with a typo, or if you accidentally selected the wrong gender or age group in the questionnaire),
- erase your personal data - to request this, please email me - meaning that your name and email address will be removed from the aforementioned list,

- object to certain or all processing of your data, e.g. if you don't want to be contacted whatsoever, you could object to my use of your email address (which I would only do for emergencies and pressing matters, this is just an example).

(This is a subset of rights guaranteed to data subjects by GDPR regulations, please refer to the disclaimer for a comprehensive overview.)

You are entitled to object to any further processing of the information we hold about you (except where it is de-identified).

You can exercise these rights or learn more about data protection in relation to this study by contacting the researcher, the PI at gareth.young@tcd.ie or the Trinity College Data Protection Officer (contact details above).

Please note that these rights relate to data which could identify you (personal data). If your data has been anonymised, we will not be able to access or delete it as we will have no way of being able to link the data to you.

If you are unhappy with how we have used your personal data, you can raise a concern with the Data Protection Commission via their online form - forms.dataprotection.ie/contact - or contact the Commission at:
Data Protection Commission
21 Fitzwilliam Square South
Dublin 2
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www.dataprotection.ie